Geometric Foundations and Structural Challenges in Neural Modeling of Lie–Poisson Spin Systems

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Abstract

This report provides a rigorous synthesis of the geometric mechanics governing 3D Heisenberg nanodot spin systems and the limitations of standard Hamiltonian Neural Networks (HNNs) in surrogating their dynamics. We demonstrate that the spin system constitutes a non-canonical Lie–Poisson system, rendering the constant symplectic assumptions of standard HNNs invalid. Furthermore, we analyze the structural challenges inherent in replacing Metropolis–Hastings Monte Carlo simulations with neural surrogates, focusing on graph-structured neighbor interactions, geometry-induced symmetry breaking in cylindrical nanodots, and the thermodynamic decoupling of the Hamiltonian from the annealing process.

Contents

1 Foundations of Hamiltonian Dynamics

1.1 Canonical Hamiltonian Systems

A canonical Hamiltonian system is defined on a symplectic manifold $(\mathcal{M}, \omega)$, where ω is a closed, non-degenerate 2-form. In a local coordinate chart $\mathbf{z} = (\mathbf{q}, \mathbf{p}) \in \mathbb{R}^{2n}$, the dynamics are dictated by a scalar Hamiltonian function $\mathcal{H} : \mathcal{M} \to \mathbb{R}$.

[Expandir la definición de symplectic manifold y el diagrama de coordenadas locales]

1.2 Symplectic Structure and Hamilton's Equations

The symplectic form ω induces a mapping between the tangent and cotangent bundles. In canonical coordinates, the evolution of the system is governed by Hamilton's equations:

$$\dot{q}_i = \frac{\mathrm{d}q_i}{\mathrm{d}t} = \frac{\partial \mathcal{H}}{\partial p_i}, \quad \dot{p}_i = \frac{\mathrm{d}p_i}{\mathrm{d}t} = -\frac{\partial \mathcal{H}}{\partial q_i} \tag{1}$$

Defining the state vector $\mathbf{z}$, these equations are compactly expressed as:

$$\dot{\mathbf{z}} = \mathbb{J}\nabla\mathcal{H}(\mathbf{z}) \tag{2}$$

where $\mathbb{J}$ is the constant, skew-symmetric Poisson tensor:

$$\mathbb{J} = \begin{pmatrix} 0 & I_n & -I_n & 0 \end{pmatrix} \tag{3}$$

[Expandir Z Punto y Tensor J]

1.3 Liouville's Theorem and Conservation Laws

Hamiltonian flow Φ_t is volume-preserving in phase space, a result known as Liouville's Theorem. Furthermore, the Hamiltonian is a conserved quantity along the flow:

$$\frac{d\mathcal{H}}{dt} = \nabla\mathcal{H} \cdot \dot{\mathbf{z}} = \nabla\mathcal{H} \cdot \mathbb{J}\nabla\mathcal{H} = 0, \tag{4}$$

due to the skew-symmetry of $\mathbb{J}$.

[Expandir Teorema de Liouville y relacionarlo con Conservación del Hamiltoniano S.Spunto]

2 Symmetries in Physical Systems

2.1 Symmetry and Noether's Theorem

A symmetry in a Hamiltonian framework is a transformation that preserves the symplectic form and leaves the Hamiltonian invariant. Noether's theorem establishes a formal correspondence between such symmetries and conserved quantities (momentum maps). If $\mathcal{H}$ is invariant under the action of a Lie group G, the associated generator F satisfies $\{F, \mathcal{H}\} = 0$, implying F is a constant of motion.

[Forma Funcional de la Invarianza y Expandir Corchete]

2.2 Rotational and Translational Symmetry

In the Heisenberg model, global rotational symmetry $SO(3)$ implies the conservation of total magnetization in the absence of external fields or anisotropy. Conversely, translational symmetry is typically present in bulk lattices but is explicitly broken in confined nanodot geometries, where the boundary $\partial\Omega$ introduces spatially varying energy contributions.

[Expandir Simetrías y su significado, ¿cómo aparecen en el nanodot?]

3 Canonical and Lie–Poisson Systems

3.1 Poisson Manifolds and State-Dependent Tensors

A Poisson manifold is characterized by a Poisson bracket $\{\cdot,\cdot\}$ that satisfies the Jacobi identity but may have a state-dependent Poisson tensor $\mathbf{B}(\mathbf{z})$:

$$\{F, G\} = \nabla F^T \mathbf{B}(\mathbf{z}) \nabla G \tag{5}$$

Unlike canonical systems where $\mathbf{B}(\mathbf{z}) = \mathbb{J}$ is constant, Lie–Poisson systems feature a tensor that depends linearly on the coordinates of the dual of a Lie algebra $\mathfrak{g}^*$.

[Expandir Identidad de Jacobi]

3.2 Non-Canonical Nature of the 2-Sphere

Individual spins $\mathbf{S}_i$ are constrained to the 2-sphere S^2. Topologically, S^2 is a symplectic manifold but does not admit global canonical coordinates $(\mathbf{q}, \mathbf{p})$. Any mapping such as $(\phi, \cos\theta)$ introduces coordinate singularities at the poles, violating the global structure required for standard canonical HNNs.

[Expandir Topología, Cambios del Espacio de Fase y Singularidades. Derivar y Explicar Mapeo 2D de Esféricas]

4 The Heisenberg Nanodot as a Lie–Poisson System

4.1 The Hamiltonian Functional

We consider N spins $\mathbf{S}_i \in \mathbb{R}^3$ with $|\mathbf{S}_i| = 1$. The total Hamiltonian is:

$$\begin{aligned}\mathcal{H}\left(\mathbf{S}|\hat{\mathbf{d}}_{ij}, \hat{\mathbf{e}}_x; \boldsymbol{\theta}\right) = &-\sum_{r=1}^{4} \tilde{J}_r \sum_{i=1}^{N} \sum_{j\in\mathcal{N}_r(i)} \langle \mathbf{s}_i, \mathbf{s}_j \rangle + \tilde{D} \sum_{i=1}^{N} \sum_{j\in\mathcal{N}_1(i)} \langle \hat{\mathbf{d}}_{ij}, (\mathbf{s}_i \times \mathbf{s}_j) \rangle \\ &- \tilde{H} \sum_{i=1}^{N} \langle \hat{\mathbf{e}}_x, \mathbf{s}_i \rangle + \frac{1}{S_o^4} \sum_{i=1}^{N} \tilde{K}_i^{eff} \left(s_{i\hat{x}}^2 s_{i\hat{y}}^2 + s_{i\hat{x}}^2 s_{i\hat{z}}^2 + s_{i\hat{y}}^2 s_{i\hat{z}}^2 \right),\end{aligned}$$

where $\tilde{J}ij$ accounts for exchange interactions up to 4th neighbors, $\tilde{\mathbf{D}}ij$ is the Dzyaloshinskii–Moriya vector, $\tilde{H}ex$ is the Zeeman field, and $\tilde{K}_i$ is the magnetocrystalline anisotropy constant.

[Explicar mejor Hamiltoniano e Interacciones]

4.2 Derivation of the Lie–Poisson Dynamics

The fundamental Lie–Poisson bracket for the $\mathfrak{so}(3)$ algebra is:

$$\{S_i^a, S_j^b\} = \epsilon^{abc} S_i^c \delta_{ij} \tag{6}$$

This induces the Poisson tensor $\mathbf{B}(\mathbf{S})$, a block-diagonal matrix where each block is the skew-symmetric matrix $\hat{\mathbf{S}}_i$:

$$\mathbf{B}_i(\mathbf{S}_i) = \begin{pmatrix} 0 & S_i^z & -S_i^y & -S_i^z & 0 & S_i^x & S_i^y & -S_i^x & 0 \end{pmatrix} \tag{7}$$

The equations of motion are given by $\dot{\mathbf{S}} = \mathbf{B}(\mathbf{S})\nabla\mathcal{H}$, yielding the torque equation:

$$\frac{d\mathbf{S}_i}{dt} = \mathbf{S}_i \times \left(-\frac{\partial \mathcal{H}}{\partial \mathbf{S}_i}\right) \tag{8}$$

This system is strictly non-canonical because the Poisson tensor $\mathbf{B}(\mathbf{S})$ is a non-constant function of the state.

[Expandir Derivación]

5 Hamiltonian Neural Networks (HNNs)

5.1 Canonical HNN Limitations

The original HNN (Greydanus et al., 2019) assumes $\dot{\mathbf{z}} = \mathbb{J}\nabla\mathcal{H}_\theta$. As established, spin systems do not possess a constant $\mathbb{J}$. Attempts to force the system into a canonical form via projection of spherical coordinates over a 2D Phase Space $(\phi, \cos\theta)$ mapping, result in:

1. Vanishing gradients near $\theta = 0, \pi$.
2. Failure to preserve the spin norm $|\mathbf{S}_i| = 1$ without soft-constraint penalties.

5.2 Generalized Lie–Poisson HNN

To respect the geometric structure of the Heisenberg system, the neural surrogate must be formulated as:

$$\dot{\mathbf{S}} = \mathbf{B}(\mathbf{S})\nabla\mathcal{H}_\theta(\mathbf{S}), \tag{9}$$

where $\mathbf{B}(\mathbf{S})$ is the hard-coded Lie–Poisson tensor. This ensures $\mathbf{S}_i \cdot \dot{\mathbf{S}}_i = 0$ by construction, preserving the manifold constraints and energy conservation.

6 Neighbor-Dependent Operations

6.1 Graph-Structured Locality

The energy contributions from exchange and DMI are inherently local but span multiple coordination shells. This defines a graph $\mathcal{G} = (V, E)$. A neural network must process these interactions via graph-structured operations to maintain efficiency.

[Explicar cómo nace la estructura de grafo y porqué es eficiente]

6.2 Representational and Algorithmic Challenges

The challenge is primarily representational. Standard MLPs cannot capture the permutation invariance and varying neighborhood sizes of a 4th-neighbor interaction without a massive increase in parameters, which leads to poor scaling for $N \approx 5000$.

[Explicar Invarianza a la Permutación y Captura de Interacciones con Kernels Convolucionales y Demás]

Guiding Research Questions

- Can Message Passing Neural Networks (MPNNs) be constrained to maintain the Lie–Poisson structure across edge updates?
- How does the truncation of the interaction kernel at the 4th neighbor affect the spectral properties of the learned Hamiltonian?

[Reconfigurar Preguntas, deben ser más generales]

7 Geometry-Aware Interactions

7.1 Cylindrical Confinement and Symmetry Breaking

The cylindrical geometry breaks translational symmetry. Surface spins experience a coordination deficit and a specific surface anisotropy $\tilde{K}_S$ directed along the radial normal $\hat{n}$.

[Expandir]

7.2 Feature Engineering vs. Geometric Inductive Biases

Boundary effects are not trivial features; they represent a fundamental change in the local Hamiltonian functional. A surrogate must distinguish between bulk and surface regimes to avoid unphysical behavior at the nanodot edges.

[Expandir]

Guiding Research Questions

- Can spatially-conditioned Hypernetworks be used to modulate the Hamiltonian parameters as a function of radial distance?
- How can rotational equivariance be preserved in the presence of radial anisotropy?

[Reconfigurar Preguntas, deben ser más generales]

8 Annealing Temperature and Sampling

8.1 Metropolis–Hastings and Thermal Fluctuations

The Metropolis–Hastings algorithm samples the Boltzmann distribution $\rho(\mathbf{S}) \propto \exp(-\beta\mathcal{H})$. The acceptance probability is:

$$P_{acc} = \min\left(1, \exp\left(-\frac{\Delta\mathcal{H}}{k_B T}\right)\right) \tag{10}$$

[Referencia y Explicación del Algoritmo]

8.2 Hamiltonian vs. Stochastic Sampling

A crucial structural realization is that T is an external parameter of the sampler, not a variable within $\mathcal{H}$. Consequently, an HNN trained to surrogate the Hamiltonian cannot, in isolation, predict the thermalization or annealing behavior of the system.

[Expandir]

Guiding Research Questions

- Is it possible to formulate a Neural Langevin Operator that surrogates the entire Markov chain transition kernel?
- How can the surrogate effectively model the critical slowing down near phase transitions without explicitly simulating the stochastic noise?

[Malas Preguntas]

9 Conclusion

The successful neural modeling of Heisenberg nanodots requires a departure from canonical Hamiltonian surrogates. By adopting a Lie–Poisson framework, we preserve the intrinsic geometry of the spin manifold. However, the replacement of stochastic sampling remains an open challenge, necessitating future research into Graph Neural Operators and thermodynamic-aware architectures that can bridge the gap between energy landscapes and statistical sampling.

[Relacionar Conclusiones, no deben hacer sugerencias]

References

[1] Greydanus, S., Dzamba, M., and Yosinski, J. (2019). Hamiltonian Neural Networks. *Advances in Neural Information Processing Systems.*

[2] Marsden, J. E., and Ratiu, T. S. (2013). *Introduction to Mechanics and Symmetry.* Springer Science Business Media.